



Figure 3: Repository selection



Figure 4: Package selection widget

the configuration for distro building. It will be processed by the *buildfarm* queue underlying in the Pardus server. You can check back after a few hours and look into the 'user log' page to get the status of the build you have requested, the link for download, the link to the project file and the Pardusman log file. Also, a history of all builds you have ever made.

Getting started

I started my work with UI (user interface) design. I worked out the basic pages and their structure. Then I designed these using Inkscape and published them on my blog for feedback. As it happens with FOSS projects in general, I too got a lot of feedback. In fact, one of my good friends (Hiran) helped me give the finishing touches to the final set of images I required for the front-end UI.

Typically, a project has several development versions of the same app while coding. Under the

circumstances, the manual way of backing up the code makes everything complex and difficult. Sometimes this even leads to loss of data/code. So, it is always advisable to use some kind of version control system (VCS) to keep track of code.

Git is one of the best version control systems available. I decided to go with Git, synced with github.com, where several FOSS projects are housed. You can pull the latest copy of Pardusman, using:

```
$ git clone git://github.com/33rm1n41/pardusman.git
```

Coming back to the UI elements, my next job was to create the set of Web pages using HTML and CSS. I scripted each page to complete the Web wizard described earlier. My template was now ready.

However, since the wizard, whose purpose is to collect data from a user, consists of several pages, it required transition from page to page. Page-to-page transitions were in traditional styles and not of Web 2.0 standards. So, I decided to use *div* containers, which can dynamically load HTML content using AJAX. This made the requirement for refreshing and loading new pages through URLs, redundant.

JQuery is a rich open source JavaScript library that helps to implement AJAX methods and calls using its rich in-built functionalities. I could easily implement the dynamic loading of pages and simple animation effects, using JQuery.

Once the templates and basic AJAX loading were completed, I started off with Django, Python's own Web framework. Django is a rapid Web application development platform. It makes it easier to develop complex Web applications in a short span of time. The coding with Pardusman progressed very fast.

The database storage in Django is handled using its own data model object-oriented structures. Initially, I wrote the pages for the user account sign up and sign in. Django was pretty easy to follow and code.

I got stuck with coding while creating a complex package selection widget. A package selection widget consists of a drop-down of packages with components in the top level and check boxes attached to it. It was solved with JQuery hacks. Thanks to the #jquery IRC channel of irc.freenode.net for helping me out.

There will be more than one package repository, like *pardus-2008* and *pardus-2009*. By selecting any one of them, we can build either a Pardus 2009 or a Pardus 2008 distribution. Dynamically producing the package information and bringing it to the packages widget causes overheads and puts a heavy load on the CPU. There might be thousands of packages in the repo and hundreds of users simultaneously accessing Pardusman, which would surely bring the server down. So I decided to build a static repo package information creator that would be scheduled to update the packages information